

Interactive Visualization of Dynamic Functional Connectivity in the ABIDE Autism Dataset

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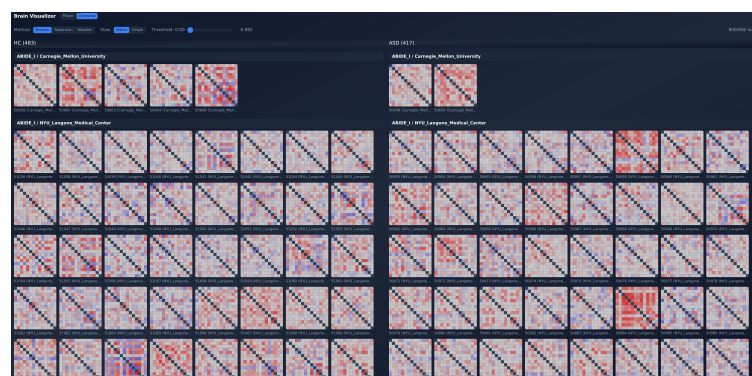


Figure 1: Overview tab: per subject connectivity matrix split into control ($N = 483$) and ASD ($N = 417$) classes grouped by acquisition site.

Abstract

Resting state functional connectivity between brain regions is increasingly understood to be a dynamic process, varying in time [HWA*13, PBVDV16]. However, the tools currently available collapse a recording into a single correlation matrix and discard the time variations. We present an interactive web application for exploring the time varying functional connectivity between brain regions in the large ABIDE I+II autism fMRI dataset. Each recording is decomposed into resting state networks whose connectivity is measured over a sliding window using several methods, including wavelet lead coherence, a directional measure of which network leads which. These connections are visualized as the edge weights of a graph, alongside a complementary overview pane to compare individual scans, or one diagnostic class against the other. This system turns a large, otherwise opaque dataset into a visual format that can be used to discover time varying functional connectivity patterns.

CCS Concepts

• Human-centered computing → Visualization; • Applied computing → Life and medical sciences

1. Introduction

Resting state fMRI (rs-fMRI) is the most studied imaging method for autism spectrum disorder (ASD) [SPAZZ24], and the functional connectivity it reveals is central to studying the condition. This connectivity is conventionally computed once over an entire recording, resulting in one static con-

nectivity matrix per subject. However, the functional connectivity varies in time and may provide additional insight into the underlying patterns, information which static correlations discard [CG09].

The Autism Brain Imaging Data Exchange (ABIDE) is a large, multisite rs-fMRI dataset. Recent work prepro-

cessed ABIDE I+II using independent component analysis (ICA) into resting state networks (RSNs) for 900 subjects (417 ASD, 483 controls) [SPA*25]. ABIDE is the most comprehensive fMRI ASD dataset [SPA24] and is an ideal dataset for this dynamic functional connectivity visualization. However, common connectivity visualizers are static and lack the interactive, time varying exploration needed to understand dynamic patterns. We present a web application that fills this gap with an animated graph view of windowed connectivity between 14 RSNs, using both symmetric correlation and wavelet lead coherence, a comparative overview of the entire diagnostic class, and time varying connectivity computed for each subject from configurable parameters.

2. Data and Connectivity

Group ICA with dual regression yields, per subject, 32 component time series, of which 14 are kept as resting state networks [SPA*25]. This results in a $T \times 14 \times 14$ correlation tensor. A window of fixed length slides across the time series data [HWA*13, PBVDV16], and at each step a 14×14 connectivity matrix is computed over all network pairs, producing a sequence of T connectivity matrices.

The system offers three correlation measures. Pearson and Spearman correlation produce symmetric matrices in the range $[-1, 1]$. The third is a directional (asymmetric) measure, called wavelet lead coherence. The wavelet transform decomposes each signal into time and frequency, capturing how its frequency content varies over time rather than averaging over the whole scan. Wavelet coherence then acts as a localized correlation coefficient between two signals, and its cross-wavelet phase angle gives their lead or lag at each point in time and frequency [GMJ04, TC98, CG09], classifying each network pair as in-phase, anti-phase, leading, or lagging [BAZ17, CPB*23]. The wavelet lead coherence measure reports values in the range $[0, 1]$, the fraction of time and frequency points at which one network leads the other, computed in both directions, so the resulting matrix is asymmetric. From this directional measure of correlation, a causality between network activations may be inferred [CPB*23].

3. Visualization

The graph view places the resting state networks on a ring and draws the current frame's connectivity matrix as edges (Figure 2). The color of an edge encodes the correlation value on a blue to red scale and thickness encodes its absolute magnitude. A configurable threshold is used to hide weak edges and denoise the visualization. For wavelet lead coherence, the edges carry arrows showing the directionality of the correlation. The frames are animated over time, showing how the graph changes. Hovering over a network node isolates its edges, and hovering over an edge shows the two networks it connects and their connectivity correlation value. This view can also play aggregate sequences, the

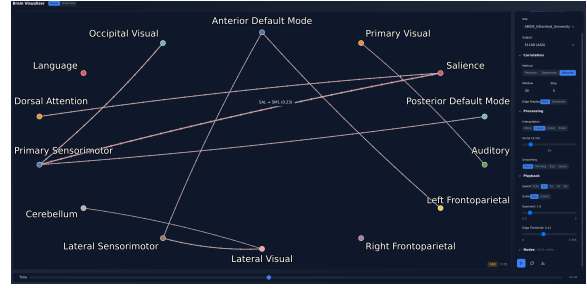


Figure 2: Graph view: 14 resting state networks on a ring with the current window's connectivity drawn as edges, where color encodes each correlation value and thickness encodes its magnitude.

mean or median over all ASD or all control subjects, so an individual's dynamics can be compared against the mean or median of either class.

A complementary overview (Figure 1) reduces each subject to a single correlation matrix over the entire recording, shown as a heatmap or a miniature graph, separated into healthy control and ASD columns and grouped by acquisition site. Read against the class mean and median, the overview enables fast scanning for individual outliers, group contrasts, and site variation. Identifying site variation and measurement error induced outliers is especially important, as it is a known confound in multisite rs-fMRI [SPA24].

The system is a web application: a TypeScript and React client renders the graph and overview on an HTML canvas. The correlation matrices are computed by a Python (FastAPI) backend service that parses the ABIDE time series and computes the connectivity correlations on demand. It is publicly available at <https://brainviz.ideabattery.org/>.

4. Conclusions

The system produces dynamic functional connectivity visualizations of the large autism fMRI dataset ABIDE I+II, enabling interactive exploration of each subject's network connectivity over time. Planned work adds a quantitative comparison of ASD against control subjects, turning exploratory observations into testable group differences and into features that future diagnostic models can be trained on.

Acknowledgements

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References

- [BAZ17] BERNAS A., ALDENKAMP A. P., ZINGER S.: Wavelet coherence-based classifier: A resting-state functional MRI study on neurodynamics in adolescents with high-functioning autism. *Computer Methods and Programs in Biomedicine* 154 (11 2017), 143–151. URL: <https://doi.org/10.1016/j.cmpb.2017.11.017>, doi:10.1016/j.cmpb.2017.11.017. 2
- [CG09] CHANG C., GLOVER G. H.: Time–frequency dynamics of resting-state brain connectivity measured with fMRI. *NeuroImage* 50, 1 (12 2009), 81–98. URL: <https://doi.org/10.1016/j.neuroimage.2009.12.011>, doi:10.1016/j.neuroimage.2009.12.011. 1, 2
- [CPB*23] CÎRSTIAN R., PILMEYER J., BERNAS A., JANSEN J. F. A., BREEUWER M., ALDENKAMP A. P., ZINGER S.: Objective biomarkers of depression: A study of Granger causality and wavelet coherence in resting-state fMRI. *Journal of Neuroimaging* 33, 3 (1 2023), 404–414. URL: <https://doi.org/10.1111/jon.13085>, doi:10.1111/jon.13085. 2
- [GMJ04] GRINSTED A., MOORE J. C., JEVREJEVA S.: Application of the cross wavelet transform and wavelet coherence to geophysical time series. *Nonlinear processes in geophysics* 11, 5/6 (11 2004), 561–566. URL: <https://doi.org/10.5194/npg-11-561-2004>, doi:10.5194/npg-11-561-2004. 2
- [HWA*13] HUTCHISON R. M., WOMELSDORF T., ALLEN E. A., BANDETTINI P. A., CALHOUN V. D., CORBETTA M., DELLA PENNA S., DUYN J. H., GLOVER G. H., GONZALEZ-CASTILLO J., HANDWERKER D. A., KEILHOLZ S., KIVINIEMI V., LEOPOLD D. A., DE PASQUALE F., SPORNS O., WALTER M., CHANG C.: Dynamic functional connectivity: Promise, issues, and interpretations. *NeuroImage* 80 (5 2013), 360–378. URL: <https://doi.org/10.1016/j.neuroimage.2013.05.079>, doi:10.1016/j.neuroimage.2013.05.079. 1, 2
- [PBVDV16] PRETI M. G., BOLTON T. A., VAN DE VILLE D.: The dynamic functional connectome: State-of-the-art and perspectives. *NeuroImage* 160 (12 2016), 41–54. URL: <https://doi.org/10.1016/j.neuroimage.2016.12.061>, doi:10.1016/j.neuroimage.2016.12.061. 1, 2
- [SPA*25] SCHIELEN S. J. C., PILMEYER J., ALDENKAMP A. P., RUIJTERS D., ZINGER S.: ICA-Based Resting-State networks obtained on large Autism FMRI dataset ABIDE. *Data* 10, 7 (7 2025), 109. URL: <https://doi.org/10.3390/data10070109>, doi:10.3390/data10070109. 2
- [SPA24] SCHIELEN S. J. C., PILMEYER J., ALDENKAMP A. P., ZINGER S.: The diagnosis of ASD with MRI: a systematic review and meta-analysis. *Translational Psychiatry* 14, 1 (8 2024), 318. URL: <https://doi.org/10.1038/s41398-024-03024-5>, doi:10.1038/s41398-024-03024-5. 1, 2
- [TC98] TORRENCE C., COMPO G. P.: A practical Guide to Wavelet analysis. *Bulletin of the American Meteorological Society* 79, 1 (1 1998), 61–78. URL: [https://doi.org/10.1175/1520-0477\(1998\)079<0061:apgtwa>2.0.co;2](https://doi.org/10.1175/1520-0477(1998)079<0061:apgtwa>2.0.co;2), doi:10.1175/1520-0477(1998)079. 2